

WorkSpace Booking System

Technical ERD Analysis — Database Analyst Report

Business Rules v1.0 | Prepared for ERD Construction

1. Entity Catalogue

The following entities are identified from the business rules. Each entity is described with its attributes, primary key, and derived field notes.

ENTITY: WORKSPACE	
Description	Represents the top-level operational unit. Stores all system-wide configuration parameters.
Primary Key	WorkspaceID (surrogate integer)
Attributes	WorkspaceID (PK) · WorkspaceName · GracePeriodMinutes · CancellationThresholdHours · DepositRatePercent · TaxRatePercent · DepositPaymentWindowMinutes
Notes	All configurable parameters (grace period, cancellation threshold, deposit rate, tax rate) belong to this entity and are stored as workspace-level settings, not hard-coded constants (BR-02-003, BR-05-005, BR-03-005, System Constraints — CONFIG).

ENTITY: ROOM_TYPE	
Description	Defines a category of room (e.g., Quiet Room, Gaming Room, Meeting Room). Owned and extended by the workspace owner.
Primary Key	RoomTypeID (surrogate integer)
Attributes	RoomTypeID (PK) · RoomTypeName · PrimaryType (ENUM: Individual Group) · BaseHourlyPrice · Capacity · IsActive
Notes	PrimaryType is a discriminator attribute. A room type cannot belong to both Individual and Group simultaneously (BR-01-001). BaseHourlyPrice and Capacity changes apply to future bookings only (BR-02-001).

ENTITY: ROOM	
Description	A physical room instance within the workspace, typed and configured by the owner.
Primary Key	RoomID (surrogate integer)
Attributes	RoomID (PK) · RoomTypeID (FK) · RoomName · Capacity · BaseHourlyPrice · IsActive
Notes	RoomName is immediately editable. Room type cannot change while active bookings exist (BR-02-002). IsActive = FALSE blocks new bookings. BaseHourlyPrice and Capacity are snapshotted at booking time (System Constraints — ENFORCE).

ENTITY: AMENITY	
Description	A feature or equipment item associated with a room (e.g., Wi-Fi, Projector, PlayStation Console).
Primary Key	AmenityID (surrogate integer)
Attributes	AmenityID (PK) · AmenityName · Description
Notes	Amenities are associated with a Room record, not with individual bookings. Each room must have at least one amenity (BR-01-002). Changes to a room's amenity list do not affect existing bookings.

ENTITY: ROOM_AMENITY (Junction / Associative)	
Description	Resolves the M:M relationship between Room and Amenity.
Primary Key	Composite: (RoomID, AmenityID)
Attributes	RoomID (FK, PK) · AmenityID (FK, PK)

ENTITY: USER (Customer)	
Description	A customer who makes individual or group bookings. Identity is defined by Name + Phone Number (natural composite key).
Primary Key	UserID (surrogate integer — recommended for referential integrity)
Attributes	UserID (PK) · FullName · PhoneNumber · Email (optional)
Business Key	(FullName + PhoneNumber) — must be unique; used for overlap prevention (BR-04-004).
Notes	A user may only hold one active individual booking at any given time. Overlap between pending and checked-in bookings for the same user must be rejected by the system.

ENTITY: INDIVIDUAL_BOOKING	
Description	Records a single-seat reservation made by a user for an individual room, either online or as a walk-in.
Primary Key	BookingID (surrogate integer)
Attributes	BookingID (PK) · UserID (FK) · RoomID (FK) · ShiftID_Creation (FK) · ShiftID_CheckIn (FK) · ShiftID_CheckOut (FK) · EmpID_Creation (FK) · EmpID_CheckIn (FK) · EmpID_CheckOut (FK) · Channel (ENUM: Online Walk-In) · Status (ENUM: Pending Checked-In Checked-Out Auto-Cancelled Cancelled) · PriceSnapshot · CreatedAt · GracePeriodExpiry · CheckInTime · CheckOutTime · TotalAmount
Notes	No end time is declared at creation. Duration = CheckOutTime – CheckInTime, computed at checkout. PriceSnapshot locks the base price at creation time. GracePeriodExpiry = CreatedAt + GracePeriodMinutes (online only). Walk-in bookings skip the Pending status and begin directly at Checked-In (BR-04-002).

ENTITY: GROUP_BOOKING	
Description	Records a full-room reservation for a group, requiring a scheduled time window and upfront deposit.
Primary Key	GroupBookingID (surrogate integer)
Attributes	GroupBookingID (PK) · RoomID (FK) · ShiftID_Creation (FK) · ShiftID_CheckIn (FK) · EmpID_Creation (FK) · EmpID_CheckIn (FK) · EmpID_CheckOut (FK) · Channel (ENUM: Online Walk-In) · Status (ENUM: Pending Confirmed Checked-In Completed Cancelled-Refunded Cancelled-Forfeited No-Show) · PriceSnapshot · ScheduledStart · ScheduledEnd · BookedDuration (derived) · DepositAmount · DepositPaid (bool) · DepositPaidAt · CancellationThresholdSnapshot · ActualCheckIn · ActualCheckOut · TotalAmount · TaxAmount · CreatedAt
Notes	Deposit = BasePrice × BookedDuration × DepositRate (25%). FinalBill = (BasePrice × BookedDuration) – Deposit + Taxes. Billing always uses BookedDuration, even if the group departs early (BR-05-004). Deposit becomes non-refundable after the cancellation threshold (BR-05-005). Room is blocked for the full booked window once Confirmed.

ENTITY: PAYMENT	
Description	Records a financial transaction linked to either an individual or group booking.
Primary Key	PaymentID (surrogate integer)
Attributes	PaymentID (PK) · BookingID (FK — individual or group) · BookingType (ENUM: Individual Group) · Amount · Method (ENUM: Cash Credit Card InstaPay) · PaymentType (ENUM: Deposit Balance Full) · PaidAt · ProcessedByEmpID (FK) · IsRefunded · RefundedAt · RefundReason
Notes	Individual: full payment at checkout. Group: deposit at booking creation, balance at checkout. Accepted methods are identical for both types (BR-03-004). Refund fields track deposit refunds on valid group cancellations (BR-03-003).

ENTITY: EMPLOYEE	
Description	A staff member responsible for processing bookings, check-ins, check-outs, and lost & found entries.
Primary Key	EmployeeID (surrogate integer)
Attributes	EmployeeID (PK) · FullName · PhoneNumber · Email · Role · IsActive

ENTITY: SHIFT	
Description	A defined time window during which an employee covers workspace operations.
Primary Key	ShiftID (surrogate integer)
Attributes	ShiftID (PK) · EmployeeID (FK) · ShiftDate · StartTime · EndTime
Notes	Shifts for the same employee on the same day must not overlap (BR-07-001). Every booking record links to the active shift at the time of creation and the employee who processed it (BR-07-002). Check-in and check-out may be

	handled by different employees; each step stores a separate employee reference (BR-07-003).
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ENTITY: LOST_FOUND	
Description	Logs items found on the premises, tracking their lifecycle from discovery to resolution.
Primary Key	ItemID (surrogate integer)
Attributes	ItemID (PK) · Description · RoomID (FK) · FoundAt (timestamp) · LoggedByEmpID (FK) · Status (ENUM: Found Stored Claimed Disposed) · ClaimedByName · ClaimedAt · DisposalReason · AuthorizingEmpID (FK)
Notes	All four mandatory fields (description, room, timestamp, employee) must be present at entry (BR-08-001). Claimed items record claimant identity and timestamp. Disposed items record reason and authorizing employee (BR-08-002).

ENTITY: PRODUCT	
Description	An item available for purchase through the workspace store, sold via online or on-site channels.
Primary Key	ProductID (surrogate integer)
Attributes	ProductID (PK) · ProductName · Description · Price · StockQuantity · IsActive
Notes	A single shared catalogue serves both online and on-site channels (BR-09-002). Products with StockQuantity = 0 or IsActive = FALSE are unavailable on both channels (BR-09-002, BR-09-004, BR-09-005).

ENTITY: STORE_ORDER	
Description	Records a customer purchase transaction from the workspace store.
Primary Key	OrderID (surrogate integer)
Attributes	OrderID (PK) · Channel (ENUM: Online On-Site) · OrderedAt · TotalAmount · PaymentMethod (ENUM: Cash Credit Card InstaPay) · UserID (FK, nullable) · CustomerName (nullable) · CustomerPhone (nullable) · ProcessedByEmpID (FK, nullable)
Notes	Customer identity is optional for guest/walk-in purchases (BR-09-003). Channel is mandatory. Store transactions are recorded independently from booking payments (BR-09-001).

ENTITY: ORDER_ITEM (Junction / Line Item)	
Description	Resolves M:M between STORE_ORDER and PRODUCT, capturing quantity and price at time of purchase.
Primary Key	Composite: (OrderID, ProductID)
Attributes	OrderID (FK, PK) · ProductID (FK, PK) · Quantity · UnitPriceSnapshot

2. Relationships & Cardinality

The table below enumerates all identified relationships between entities, with explicit cardinality notation.

Entity A	Entity B	Cardinality	Description
WORKSPACE	ROOM_TYPE	1 : M	A workspace defines many room types; each room type belongs to exactly one workspace.
ROOM_TYPE	ROOM	1 : M	A room type categorises many rooms; each room has exactly one type.
ROOM	AMENITY	M : M	A room has one or more amenities; an amenity may be assigned to many rooms. Resolved by ROOM_AMENITY junction.
ROOM	INDIVIDUAL_BOOKING	1 : M	An individual room can have many bookings over time; each individual booking targets exactly one room.
ROOM	GROUP_BOOKING	1 : M	A group room can have many bookings over time; each group booking targets exactly one room.
USER	INDIVIDUAL_BOOKING	1 : M	A user may hold many individual bookings; each individual booking belongs to exactly one user.
INDIVIDUAL_BOOKING	PAYMENT	1 : M	A booking can have one or more payment records (e.g., partial installment or refund record).
GROUP_BOOKING	PAYMENT	1 : M	A group booking can have multiple payment records: deposit and balance settlement.
EMPLOYEE	SHIFT	1 : M	An employee works many shifts; each shift is assigned to exactly one employee.
SHIFT	INDIVIDUAL_BOOKING	1 : M	Each booking creation event is attributed to an active shift. A shift may cover many booking events.
SHIFT	GROUP_BOOKING	1 : M	Group booking creation is linked to the active shift at time of booking.
EMPLOYEE	INDIVIDUAL_BOOKING	1 : M (×3)	Three separate employee FKs on each booking: EmpID_Creation, EmpID_CheckIn, EmpID_CheckOut — an employee may handle many booking events.
EMPLOYEE	GROUP_BOOKING	1 : M (×3)	Same three-role employee traceability as individual bookings (BR-07-003).

ROOM	LOST_FOUND	1 : M	A room may have many lost & found items logged against it; each item is found in one room.
EMPLOYEE	LOST_FOUND	1 : M (×2)	LoggedByEmpID and AuthorizingEmpID: an employee may log or authorize many lost & found entries.
STORE_ORDER	PRODUCT	M : M	An order may contain many products; a product may appear in many orders. Resolved by ORDER_ITEM junction.
USER	STORE_ORDER	1 : M (optional)	A registered user may place many orders; a store order may optionally be linked to a user (guest orders: nullable FK).

3. Constraints & Business Policies

All constraints are derived directly from the business rules and must be enforced at the database or application layer.

Rule ID	Type	Description
BR-01-001	CHECK	Room PrimaryType must be exactly one of {Individual, Group}. No NULL or dual-type values permitted.
BR-01-002	CHECK	Each room must have at least one entry in the ROOM_AMENITY junction table.
BR-02-001	BUSINESS	Price and Capacity changes are prospective only. Existing Pending, Confirmed, and Checked-In bookings retain the snapshot values captured at creation.
BR-02-002	BLOCK	Room PrimaryType cannot be modified while any booking with Status IN (Pending, Confirmed, Checked-In) exists for that room.
BR-02-003	CONFIG	Grace period (in minutes) is a workspace-level parameter. Changes apply only to newly created bookings.
BR-03-001	ENFORCE	Group booking status remains Pending until the 25% deposit payment is recorded.
BR-03-002	BUSINESS	Deposit is non-refundable in the event of early group departure. Billing always uses BookedDuration.
BR-03-003	BUSINESS	Deposit refund is conditional on cancellation occurring before the CancellationThreshold. No-shows forfeit the deposit unconditionally.
BR-04-004	BLOCK	No two active individual bookings (Pending or Checked-In) for the same user (Name + Phone) may have overlapping time ranges.
BR-05-002	BLOCK	No two Confirmed group bookings for the same room may have overlapping time windows. Overlap: start1 < end2 AND start2 < end1.
BR-05-003	ENFORCE	Group bookings not checked in by ScheduledStart are automatically marked No-Show. Deposit is forfeited and room is released.

BR-05-007	BLOCK	Partial group room bookings are prohibited. A group booking reserves the entire room for the full booked duration.
BR-06-001	ENFORCE	Availability is recalculated in real time on every booking state change.
BR-06-003	BLOCK	New bookings are rejected when AvailableSeats = 0 (individual) or any Confirmed booking overlaps the requested window (group).
BR-06-004	ENFORCE	Pending individual bookings within their grace window reduce available seat count. Expired pending bookings do not count.
BR-07-001	CHECK	No two shifts for the same employee on the same calendar date may have overlapping start/end times.
BR-07-002	AUDIT	Every booking record must reference the active ShiftID and EmployeeID at the time of booking creation and check-in.
BR-08-001	NOT NULL	LOST_FOUND entries must have: Description, RoomID, FoundAt, LoggedByEmpID. All four fields are mandatory.
BR-09-002	CHECK	Products with StockQuantity = 0 or IsActive = FALSE must not be purchasable through either channel.
BR-09-005	ENFORCE	StockQuantity is shared across channels and must be decremented atomically on each sale to prevent overselling.
SYS-01	AUDIT	All booking status transitions, payment records, price overrides, and lost & found entries must be timestamped and employee-attributed.
SYS-02	ENFORCE	PriceSnapshot and CapacitySnapshot on each booking record the values at the moment of creation and cannot be retroactively modified.

4. Enumerated Types Summary

The following ENUM domains are used across entities to enforce controlled vocabularies at the data-model level.

Attribute / Context	Allowed Values
Room PrimaryType	Individual Group
Individual Booking Status	Pending Checked-In Checked-Out Auto-Cancelled Cancelled
Group Booking Status	Pending Confirmed Checked-In Completed Cancelled-Refunded Cancelled-Forfeited No-Show
Booking Channel	Online Walk-In
Payment Method	Cash Credit Card InstaPay
Payment Type	Deposit Balance Full
Lost & Found Status	Found Stored Claimed Disposed
Store Order Channel	Online On-Site

5. ERD Construction Notes

5.1 Supertype / Subtype Pattern (Booking)

INDIVIDUAL_BOOKING and GROUP_BOOKING are distinct entity types sharing the conceptual 'Booking' supertype. They differ significantly in attributes, lifecycle states, billing formulas, and channel behaviour. A shared BOOKING supertype table with a discriminator column is one valid physical implementation; alternatively, two fully separate tables avoid NULL-heavy rows and enforce their distinct constraints cleanly.

5.2 Three-Role Employee Pattern

Both INDIVIDUAL_BOOKING and GROUP_BOOKING carry three separate employee foreign keys (EmpID_Creation, EmpID_CheckIn, EmpID_CheckOut). These are three separate 1:M relationships from EMPLOYEE to the booking table, represented as three distinct FK columns. In the ERD, draw three separate lines from EMPLOYEE to each booking entity, labelled by role.

5.3 Price & Capacity Snapshot Enforcement

The business rules mandate that price and capacity at booking creation are locked. The ERD should reflect PriceSnapshot and CapacitySnapshot as stored attributes on each booking entity rather than derived values, to honour the audit requirement and the immutability constraint.

5.4 Availability as a Derived Concept

AvailableSeats for individual rooms and RoomAvailable for group rooms are computed values, not stored attributes. They must not appear as persistent columns in the ERD. They are business-logic derivations, optionally materialised as views: $\text{AvailableSeats} = \text{Capacity} - \text{COUNT}(\text{Checked-In}) - \text{COUNT}(\text{Pending within grace period})$.

5.5 Store Independence

STORE_ORDER and ORDER_ITEM form a self-contained sub-system. The only optional link to the booking domain is UserID on STORE_ORDER (nullable FK). The store entities are drawn as a separate cluster in the ERD, connected to USER by a dashed optional relationship line.